

MEET PITTSBURNT

A CRASH TEST FOR CITIES

HackCMU 2026 - Build Brief + Data Stack

Core question

Cars are crash-tested before people drive them. Buildings are stress-tested before people enter them. What if we could stress-test Pittsburgh before residents experience the next extreme heatwave?

Select city
state



Run heat
crash test



Find human
hotspots



Adapt city



Re-test

Winning demo loop

Oakland + 3 PM + 2035 Heatwave + vulnerable population → RUN CRASH TEST → red exposure hotspots → \$250K ADAPT PITTSBURGH → add trees/shade → RUN AGAIN → modeled exposure drops.

Build rule: If the before → adapt → after loop works reliably, the MVP is done.

1. Product flow - what the demo screen needs

Product thesis

Pittsburnt does not ask “Where is Pittsburgh hottest?” It asks “Where do people accumulate the most heat while moving through the city, and what is the highest-impact intervention per dollar?”

1. MAP / SCOPE	Pittsburgh overview → zoom to Oakland MVP zone → Forbes / Fifth / Craig corridors → 50-150m segments.
2. TIME	8 AM / 12 PM / 3 PM / 6 PM buttons first. Each time changes sun position, building shadows and synthetic pedestrian demand.
3. EXISTING PROTECTION	Toggle baseline layers: tree canopy / city trees, building shade, transit shelters, cooling centers.
4. WHO	All / Students / Older adults / Workers / Mobility-constrained. Each creates a synthetic trip population with origin, destination, walking speed and departure time.
5. FUTURE SCENARIO	TODAY / 2035 HEATWAVE / 2050 HEATWAVE. Use climate projection periods rather than claiming an exact future-day temperature.
6. CRASH TEST	Run the same people through the same urban geometry under the selected climate/time condition. Accumulate exposure along each segment.
7. HOTSPOTS	Color segments by modeled human exposure: low → elevated → high → hotspot. Red means high exposure burden, not simply highest air temperature.
8. ADAPT	Choose budget, then add or optimize Trees / Shade structures / Cooling stations. Re-run identical scenario and compare before vs after.

Hero demo



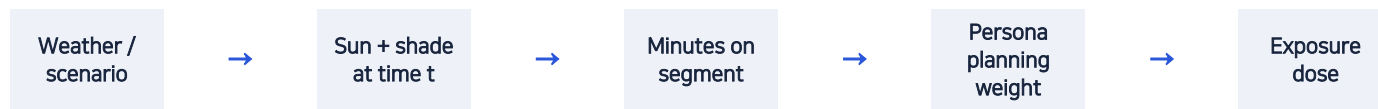
→ hotspots appear → ADAPT PITTSBURGH (\$250K) → re-test → 38% becomes 15% (illustrative until model is calibrated)

2. Exposure engine - keep it defensible and hackathon-sized

Important framing

Do not claim a medical “heatstroke probability.” Call the output a modeled Heat Exposure Score or At-risk Pedestrian Minutes. The goal is relative comparison between the same city before and after an intervention.

Segment-level calculation



MVP formula: $\text{Exposure} = \text{heat severity} \times \text{sun exposure} \times \text{time spent} \times \text{planning weight}$

Heat severity: start with NWS Heat Index (temperature + humidity). Heat Index is a shade-based measure; direct sunshine can increase heat burden, which is why the shade layer matters.

Sun exposure(segment, time): 0–1. It changes at 8 / 12 / 3 / 6 as building and tree shadows move.

Time spent: segment length ÷ walking speed. This naturally gives slower walkers more accumulated exposure without inventing medical multipliers.

Planning weight: optional city-priority weight for a vulnerable population. Label it as a planning preference, not a physiological risk coefficient.

Segment	People/hr	Sun @ 3PM	Avg minutes	Exposure priority
Forbes A	280	0.92	2.8	HIGH
Fifth B	160	0.35	2.1	MED
Craig C	220	0.76	1.7	HIGH

Recommended judge line: “The hottest street is not necessarily the street that needs shade most. We prioritize where heat, sun exposure, pedestrian volume and vulnerability overlap.”

Future climate scenario

2035 Heatwave = a stress-test value derived from the USGS CMIP6-LOCA2 2025-2049 distribution (for example, a selected hot percentile), not an exact prediction for calendar year 2035.

2050 Heatwave = same idea using the 2050-2074 projection period.

Keep the geometry and population constant when comparing TODAY vs future scenario so judges can see the climate effect cleanly.

3. 24-hour build timeline + shipping checklist

Time	Goal	Exit criterion
H0-1	Lock scope	Oakland only; demo script; one metric; three interventions.
H1-3	Map + walking graph	Map renders; corridors/segments exist; routes can be generated.
H1-3 parallel	Simulation skeleton	100+ dots/trips move or can be replayed deterministically.
H3-6	Time + shade	8/12/3/6 states; sun/shade changes; baseline protection layer appears.
H6-8	Exposure engine	Each trip gets a score; segment hotspots can be ranked.
H8-10	Interventions	Tree and shade structure alter segment exposure.
H10-12	END-TO-END MVP	Crash Test → hotspot → Adapt → re-test works without manual intervention.
H12-15	Optimizer	Greedy exposure-reduction-per-cost placement.
H15-18	Personas + climate polish	3 persona types; 2035/2050 scenario labels; assumptions panel.
H18-21	Reliability	Reset; fallback JSON; API error handling; run demo 5 times.
H21-24	Pitch / submit	No new features. Video, Devpost, architecture diagram, rehearsal.

Must-have before submission

- | | |
|--|--|
| ☐ Oakland map + segment geometry | ☐ Exposure Score + hotspot map |
| ☐ 4 time states | ☐ Tree + shade intervention |
| ☐ existing tree/building shade baseline | ☐ Budget constraint |
| ☐ 100-500 synthetic trips | ☐ Before / After metric |
| ☐ 2-3 persona types | ☐ Reset/fallback mode |
| ☐ 2035 Heatwave scenario | ☐ Assumptions + "not a medical forecast" disclaimer |

4. Data & site stack - P0 required

Priority rule: P0 blocks the core demo. P1 materially improves credibility. P2 is optional research-grade enrichment.

Pri	Source / site	What to use it for	Hackathon caveat	Link
P0	OSMnx / OpenStreetMap	Walking network; corridor edges; shortest paths; segment graph.	Fastest route to a custom pedestrian graph. Cache the Oakland graph locally.	Open
P0	Mapbox GL JS + Directions	Map UI, route display, walking routes if you do not build all routing yourself.	Requires token. Use Mapbox for visual layer even if exposure logic runs in Python.	Open
P0	City of Pittsburgh Trees (WPRDC)	Existing managed tree points; baseline protection markers.	Dataset notes it has not updated since 2020. Use as prototype baseline, not "current exact inventory."	Open
P0	Tree Pittsburgh Urban Tree Canopy	Canopy coverage baseline and 2020 canopy context.	Good for coverage context; 2020 vintage. Use point trees for quick MVP or canopy polygons/raster if accessible.	Open
P0	Allegheny County Building Footprints (WPRDC)	Building polygons for shadow projection.	The source dataset is historical; acceptable for Oakland prototype geometry with a visible limitation note.	Open
P0	SunCalc or pvlib solar position	Sun altitude + azimuth at 8/12/3/6; derive shadow direction and length.	SunCalc is easiest in JS; pvlib is stronger if your simulation is Python-first.	Open
P0	USGS CMIP6-LOCA2	Future heat stress-test scenarios for 2025-2049 and 2050-2074.	Do not label a percentile as an exact 2035 temperature. Pick scenario + percentile and disclose it.	Open
P0	NWS Heat Index / WBGT guidance	Baseline heat calculation and rationale for sun vs shade.	Heat Index is shade-based; direct sun can add substantial burden. For MVP use as relative score, not medical prediction.	Open

5. Data & site stack - P1 credibility / P2 optional

Pri	Source / site	What to use it for	Hackathon caveat	Link
P1	PRT Stops (WPRDC)	Transit stops + shelter-related stop attributes for existing shade infrastructure.	Current schedule-period dataset, republished regularly. Useful for Oakland bus-stop exposure.	Open
P1	PRT Developer Resources	GTFS / GTFS-RT / GIS if you want trips or waiting populations tied to transit.	Not required for core MVP. Add only after Crash Test loop works.	Open
P1	City Cooling Center announcements	Existing cooling-center locations and activation logic during hot periods.	Centers are activated episodically; treat as safe destinations, not permanent shade infrastructure.	Open
P1	U.S. Census ACS 5-year API	Age, disability, commute and socioeconomic distributions to calibrate synthetic personas.	Use neighborhood/block-group aggregates, not individual-level claims.	Open
P1	Census LEHD / LODES	Workplace and home-to-work patterns for worker flows and daytime population.	Best for worker personas; more preprocessing than ACS.	Open
P1	Allegheny County Property Assessments	Residential STORIES field as one possible building-height proxy.	STORIES applies to dwelling characteristics; weak for Oakland institutional/commercial buildings. Use only as supplementary height data.	Open
P2	Envision Downtown pedestrian counts	Historic hourly pedestrian/bike counts at 16 Downtown locations; useful to calibrate time-of-day flow patterns.	2016 Downtown data, so use as pattern reference, not current Oakland counts.	Open
P2	USGS 3DEP / LidarExplorer	Terrain / lidar if you want higher-fidelity elevation or building/vegetation geometry.	Powerful but too heavy for first 12 hours. Use only if the team already knows geospatial processing.	Open
P2	NWS Weather API	Live/forecast weather if you want "Today" to use a real weather state.	No need for future scenarios; CMIP6 is separate.	Open

6. Shadow implementation - do not let building height block the hack

The hard part

Building footprints are easy to obtain; reliable building heights are not. For a 24-hour hack, use a tiered fallback so moving shadows never become the critical blocker.

Tier 1 - fastest

Use OSM building:height / building:levels where available. Convert levels to an approximate height. Manually patch 10-20 hero buildings in the Oakland demo corridor if needed.

Tier 2 - supplemental

Join residential parcels to Allegheny County Property Assessments STORIES. Good for houses; do not assume it covers institutional/commercial buildings well.

Tier 3 - research-grade

Use 3DEP lidar / point cloud-derived surfaces for better height and vegetation geometry. This is future work unless someone on the team already has the pipeline.

Shadow math for MVP: $\text{shadow length} = \text{building height} / \tan(\text{solar altitude})$

Get solar altitude + azimuth from SunCalc/pvlib for Pittsburgh and the selected timestamp.

Project a simple shadow polygon from each hero building. You only need 8 / 12 / 3 / 6 snapshots for the MVP.

Intersect each street segment with shadow polygons to produce sun_exposure(segment, time).

Tree intervention can simply reduce sun_exposure on nearby affected segments. You do not need to model canopy thermodynamics.

Recommended technical split

Layer	Fast implementation	Owner
Frontend / map	Next.js + Mapbox GL JS	Frontend
Street graph	OSMnx + NetworkX	Simulation
Geospatial ops	GeoPandas + Shapely	Simulation/Data
Sun position	SunCalc (JS) or pvlib (Python)	Simulation
Exposure engine	Python / NumPy / Pandas	Simulation
Optimizer	Greedy exposure-reduction-per-cost	Algorithm
API	FastAPI + static fallback JSON	Backend

7. First 3 hours - exact implementation order

0-30 min	Freeze Oakland bbox + three corridors. Download/cache OSM walking graph. Decide the exact demo date/time and scenario labels.
30-60 min	Render Mapbox/OSM map and one synthetic route. Split route edges into segment records with IDs.
60-90 min	Load City Trees + building footprint GeoJSON clipped to the Oakland bbox. Do not process the whole county.
90-120 min	Get sun azimuth/altitude for 8/12/3/6. Make one building cast a visible shadow; then batch the hero buildings.
120-150 min	Generate 100 synthetic trips with deterministic seed. Attach walking speed/persona and route segments.
150-180 min	Compute a basic exposure score and render segment heat colors. At H3 you should already have a crude Crash Test.

Fallback rules

If tree canopy polygons are hard to obtain: use City Trees points with a fixed canopy radius for the demo.

If building heights are incomplete: patch only the visible hero corridor buildings and label heights as prototype estimates.

If future climate files are too slow to process: precompute one Oakland scenario value from the USGS period and store it in a scenario JSON with source metadata.

If live APIs fail: all core demo data should have a local cached GeoJSON/JSON fallback.

If the moving-shadow animation is unstable: keep 8/12/3/6 snapshot buttons. Do not sacrifice the Crash Test → Adapt → Re-test loop for animation.

Final judge message

"We are not forecasting a person's medical outcome. We are using published climate and city data to stress-test where modeled pedestrian heat exposure accumulates, then comparing which interventions reduce that exposure most efficiently."

Source note: Dataset availability and metadata checked September 11, 2026. Some Pittsburgh baseline datasets are intentionally older (e.g., City Trees and 2020 canopy); the prototype should expose vintage/limitations rather than imply real-time completeness.